

Derivation of the Treynor-Black Model

$$S_P^2 = S_M^2 + \frac{\alpha_A^2}{\sigma^2(e_A)}$$

Prove:

Denote the excess as $R_i (= r_i - r_f)$. From single-index model assumption,

$$R_i = \alpha_i + \beta_i R_M + e_i$$

$$\Rightarrow R_A = \alpha_A + \beta_A R_M + e_A$$

$$E(R_A) = \alpha_A + \beta_A E(R_M) \quad \because E[e_A] = 0$$

$$\sigma_A^2 = \beta_A^2 \sigma_M^2 + \sigma^2(e_A) \quad \text{and} \quad \text{COV}(R_A, R_M) = \text{COV}(\beta_A R_M, R_M) = \beta_A \sigma_M^2$$

To simplify the notation, denote $\bar{R}_A \equiv E(R_A) = \alpha_A + \beta_A \bar{R}_M$, $W_A \equiv W_A^*$, and $W_0 \equiv W_A^0$.

One can use the results of the optimization problem of two risky assets (assets 1 and 2) and riskfree asset to derive the optimal weight of the Active portfolio W_A .

$$w_1 = \frac{[\mu_1 - r_f] \sigma_2^2 - [\mu_2 - r_f] \sigma_{12}}{[\mu_2 - r_f] \sigma_1^2 + [\mu_1 - r_f] \sigma_2^2 - [\mu_1 + \mu_2 - 2r_f] \sigma_{12}}$$

$$w_2 = 1 - w_1 \tag{1}$$

From the results of Equation (1),

$$\begin{aligned} W_A &= \frac{\bar{R}_A \sigma_M^2 - \bar{R}_M \text{COV}(R_A, R_M)}{\bar{R}_A \sigma_M^2 + \bar{R}_M \sigma_A^2 - (\bar{R}_A + \bar{R}_M) \text{COV}(R_A, R_M)} \\ &= \frac{(\alpha_A + \beta_A \bar{R}_M) \sigma_M^2 - \bar{R}_M \beta_A \sigma_M^2}{(\alpha_A + \beta_A \bar{R}_M) \sigma_M^2 + \bar{R}_M (\beta_A^2 \sigma_M^2 + \sigma^2(e_A)) - (\alpha_A + \beta_A \bar{R}_M + \bar{R}_M) \beta_A \sigma_M^2} \\ &= \frac{\alpha_A \sigma_M^2}{\alpha_A \sigma_M^2 + \beta_A \bar{R}_M \sigma_M^2 + \bar{R}_M \beta_A^2 \sigma_M^2 + \bar{R}_M \sigma^2(e_A) - \alpha_A \beta_A \sigma_M^2 - \beta_A^2 \sigma_M^2 \bar{R}_M - \bar{R}_M \beta_A \sigma_M^2} \\ &= \frac{\alpha_A \sigma_M^2}{\alpha_A \sigma_M^2 + \bar{R}_M \sigma^2(e_A) - \alpha_A \beta_A \sigma_M^2} \\ &= \frac{\alpha_A \sigma_M^2}{\bar{R}_M \sigma^2(e_A) + (1 - \beta_A) \alpha_A \sigma_M^2} \\ &= \frac{\alpha_A \sigma_M^2}{\bar{R}_M \sigma^2(e_A)} \\ &= \frac{1 + (1 - \beta_A) \frac{\alpha_A \sigma_M^2}{\bar{R}_M \sigma^2(e_A)}}{\bar{R}_M \sigma^2(e_A)} \end{aligned}$$

Let $W_0 = \frac{\alpha_A / \sigma^2(e_A)}{\bar{R}_M / \sigma_M^2},$

$$W_A = \frac{\frac{\alpha_A \sigma_M^2}{\bar{R}_M \sigma^2(e_A)}}{1 + (1 - \beta_A) \frac{\alpha_A \sigma_M^2}{\bar{R}_M \sigma^2(e_A)}} = \frac{W_0}{1 + (1 - \beta_A) W_0}$$

where

$\sigma(e_i)$ = residual standard deviation of the i^{th} security (or portfolio)

To obtain the results of the Treynor–Black model as the following

$$S_p^2 = \frac{\left[W_A \cdot (\alpha_A + \beta_A \bar{R}_M) + (1 - W_A) \bar{R}_M \right]^2}{W_A^2 (\beta_A^2 \cdot \sigma_M^2 + \sigma^2(e_A)) + (1 - W_A)^2 \sigma_M^2 + 2W_A (1 - W_A) \beta_A \sigma_M^2} = S_M^2 + \frac{\alpha_A^2}{\sigma^2(e_A)},$$

one needs to first rewrite $W_A = \frac{W_0}{1 + (1 - \beta_A) W_0}$

$$\Rightarrow W_A = \frac{1}{\frac{1}{W_0} + (1 - \beta_A)} = \frac{1}{\frac{\bar{R}_M \sigma^2(e_A)}{\alpha_A \sigma_M^2} + (1 - \beta_A)} = \frac{\alpha_A \sigma_M^2}{\bar{R}_M \sigma^2(e_A) + (1 - \beta_A) \alpha_A \sigma_M^2}$$

and $(1 - W_A) = \frac{\bar{R}_M \sigma^2(e_A) - \beta_A \alpha_A \sigma_M^2}{\bar{R}_M \sigma^2(e_A) + (1 - \beta_A) \alpha_A \sigma_M^2}$

Step 1: Plugging W_A and $(1 - W_A)$ into the numerator of S_p^2

$$\begin{aligned} & \left[W_A \cdot (\alpha_A + \beta_A \bar{R}_M) + (1 - W_A) \bar{R}_M \right]^2 \\ &= \frac{\left[\alpha_A \sigma_M^2 \cdot (\alpha_A + \beta_A \bar{R}_M) + (\bar{R}_M \sigma^2(e_A) - \beta_A \alpha_A \sigma_M^2) \cdot \bar{R}_M \right]^2}{\left[\bar{R}_M \sigma^2(e_A) + (1 - \beta_A) \alpha_A \sigma_M^2 \right]^2} \\ &= \frac{\left[\alpha_A^2 \sigma_M^2 + \alpha_A \beta_A \bar{R}_M \sigma_M^2 + \bar{R}_M^2 \sigma^2(e_A) - \beta_A \alpha_A \sigma_M^2 \bar{R}_M \right]^2}{\left[\bar{R}_M \sigma^2(e_A) + (1 - \beta_A) \alpha_A \sigma_M^2 \right]^2} \\ &= \frac{\left[\alpha_A^2 \sigma_M^2 + \bar{R}_M^2 \sigma^2(e_A) \right]^2}{\left[\bar{R}_M \sigma^2(e_A) + (1 - \beta_A) \alpha_A \sigma_M^2 \right]^2} \end{aligned}$$

Step 2: Plugging W_A and $(1-W_A)$ into the denominator of S_P^2 .

$$\begin{aligned}
& W_A^2 (\beta_A^2 \cdot \sigma_M^2 + \sigma^2(e_A)) + (1-W_A)^2 \sigma_M^2 + 2W_A(1-W_A) \beta_A \sigma_M^2 \\
&= \frac{(\alpha_A \sigma_M^2)^2}{(\bar{R}_M \sigma^2(e_A) + (1-\beta_A) \alpha_A \sigma_M^2)^2} \cdot (\beta_A^2 \cdot \sigma_M^2 + \sigma^2(e_A)) \\
&\quad + \frac{(\bar{R}_M \sigma^2(e_A) - \beta_A \alpha_A \sigma_M^2)^2}{(\bar{R}_M \sigma^2(e_A) + (1-\beta_A) \alpha_A \sigma_M^2)^2} \cdot \sigma_M^2 \\
&\quad + 2 \times \frac{(\alpha_A \sigma_M^2) \cdot (\bar{R}_M \sigma^2(e_A) - \beta_A \alpha_A \sigma_M^2)}{(\bar{R}_M \sigma^2(e_A) + (1-\beta_A) \alpha_A \sigma_M^2)^2} \times \beta_A \sigma_M^2 \\
&= \frac{[\alpha_A^2 \beta_A^2 \sigma_M^6 + \alpha_A^2 \sigma_M^4 \sigma^2(e_A) + \bar{R}_M^2 \sigma^4(e_A) \sigma_M^2 - 2 \cdot \bar{R}_M \sigma^2(e_A) \beta_A \alpha_A \sigma_M^4 + \alpha_A^2 \beta_A^2 \sigma_M^6 + 2 \cdot \bar{R}_M \sigma^2(e_A) \beta_A \alpha_A \sigma_M^4 - 2 \cdot \alpha_A^2 \beta_A^2 \sigma_M^6]}{(\bar{R}_M \sigma^2(e_A) + (1-\beta_A) \alpha_A \sigma_M^2)^2} \\
&= \frac{(\alpha_A^2 \sigma_M^4 \sigma^2(e_A) + \bar{R}_M^2 \sigma^2(e_A) \sigma_M^2)}{(\bar{R}_M \sigma^2(e_A) + (1-\beta_A) \alpha_A \sigma_M^2)^2} \\
&= \frac{\sigma_M^2 \sigma^2(e_A) \cdot (\alpha_A^2 \sigma_M^2 + \bar{R}_M^2 \sigma^2(e_A))}{(\bar{R}_M \sigma^2(e_A) + (1-\beta_A) \alpha_A \sigma_M^2)^2}
\end{aligned}$$

Therefore

$$\begin{aligned}
S_P^2 &= \frac{[\alpha_A^2 \sigma_M^2 + \bar{R}_M^2 \sigma^2(e_A)]^2}{[\bar{R}_M \sigma^2(e_A) + (1-\beta_A) \alpha_A \sigma_M^2]^2} \times \frac{(\bar{R}_M \sigma^2(e_A) + (1-\beta_A) \alpha_A \sigma_M^2)^2}{\sigma_M^2 \sigma^2(e_A) \cdot (\alpha_A^2 \sigma_M^2 + \bar{R}_M^2 \sigma^2(e_A))} \\
&= \frac{\alpha_A^2 \sigma_M^2 + \sigma^2(e_A) \bar{R}_M^2}{\sigma_M^2 \sigma^2(e_A)} = \frac{\alpha_A^2}{\sigma^2(e_A)} + \frac{\bar{R}_M^2}{\sigma_M^2} = \frac{\alpha_A^2}{\sigma^2(e_A)} + S_M^2
\end{aligned}$$

Proof of Equation (1)

$\therefore$ risky portfolio $\therefore w_1 + w_2 = 1$

$$\theta_p = \frac{\mu_p - r_f}{\sigma_p} = \frac{w_1\mu_1 + (1-w_1)\mu_2 - r_f}{\left[w_1^2\sigma_1^2 + (1-w_1)^2\sigma_2^2 + 2w_1(1-w_1)\sigma_{12} \right]^{1/2}}$$

$$\frac{d\theta_p}{dw_1} = \frac{(\mu_1 - \mu_2) \left(w_1^2\sigma_1^2 + (1-w_1)^2\sigma_2^2 + 2w_1(1-w_1)\sigma_{12} \right)}{\left[w_1^2\sigma_1^2 + (1-w_1)^2\sigma_2^2 + 2w_1(1-w_1)\sigma_{12} \right]^{3/2}}$$

$$- \frac{(w_1\mu_1 + \mu_2 - \mu_2w_1 - r_f) \left(w_1\sigma_1^2 - \sigma_2^2 + w_1\sigma_2^2 + \sigma_{12} - 2w_1\sigma_{12} \right)}{\left[w_1^2\sigma_1^2 + (1-w_1)^2\sigma_2^2 + 2w_1(1-w_1)\sigma_{12} \right]^{3/2}}$$

Let $\frac{d\theta_p}{dw_1} = 0$

$$\Rightarrow 0 = \mu_1\sigma_1^2w_1^2 + \mu_1\sigma_2^2 - 2\mu_1\sigma_2^2w_1 + \mu_1\sigma_2^2w_1^2 + 2\mu_1\sigma_{12}w_1 - 2\mu_1\sigma_{12}w_1^2$$

$$- \mu_2\sigma_1^2w_1^2 - \mu_2\sigma_2^2 + 2\mu_2\sigma_2^2w_1 - \mu_2\sigma_2^2w_1^2 - 2\mu_2\sigma_{12}w_1 + 2\mu_2\sigma_{12}w_1^2$$

$$- (\mu_1\sigma_1^2w_1^2 - \mu_1\sigma_2^2w_1 + \mu_1\sigma_2^2w_1^2 + \mu_1\sigma_{12}w_1 - 2\mu_1\sigma_{12}w_1^2)$$

$$- (\mu_2\sigma_1^2w_1 - \mu_2\sigma_2^2 + \mu_2\sigma_2^2w_1 + \mu_2\sigma_{12} - 2\mu_2\sigma_{12}w_1)$$

$$+ (\mu_2\sigma_1^2w_1^2 - \mu_2\sigma_2^2w_1 + \mu_2\sigma_2^2w_1^2 + \mu_2\sigma_{12}w_1^2 - 2\mu_2\sigma_{12}w_1^2)$$

$$+ r_f (\sigma_1^2w_1 - \sigma_2^2 + \sigma_2^2w_1 + \sigma_{12} - 2\sigma_{12}w_1)$$

$$\Rightarrow 0 = r_f (\sigma_1^2 + \sigma_2^2 - 2\sigma_{12})w_1 + (\mu_1\sigma_{12} - \mu_1\sigma_2^2 - \mu_2\sigma_1^2 + \mu_2\sigma_{12})w_1$$

$$+ r_f (\sigma_{12} - \sigma_2^2) + \mu_1\sigma_2^2 - \mu_2\sigma_{12}$$

$$\Rightarrow w_1 = \frac{r_f (\sigma_2^2 - \sigma_{12}) + \mu_2\sigma_{12} - \mu_1\sigma_2^2}{r_f (\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}) + \mu_1 (\sigma_{12} - \sigma_2^2) + \mu_2 (\sigma_{12} - \sigma_1^2)}$$

$$w_2 = 1 - w_1$$

Or

$$\Rightarrow w_1 = \frac{[r_f - \mu_1]\sigma_2^2 - [r_f - \mu_2]\sigma_{12}}{[r_f - \mu_2]\sigma_1^2 + [r_f - \mu_1]\sigma_2^2 + [\mu_1 + \mu_2 - 2r_f]\sigma_{12}}$$

$$\Rightarrow w_1 = \frac{[\mu_1 - r_f]\sigma_2^2 - [\mu_2 - r_f]\sigma_{12}}{[\mu_2 - r_f]\sigma_1^2 + [\mu_1 - r_f]\sigma_2^2 - [\mu_1 + \mu_2 - 2r_f]\sigma_{12}}$$

$$w_2 = 1 - w_1$$